The Hillingdon Hospitals
NHS Foundation Trust

Smoking Cessation and Carbon Monoxide Monitoring in Maternity

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Change History

Version	Author	Summary of changes
2.0	Gillian Pearce	• Removal of Covid 19 Pandemic • Additional information about process following referral to CNWL for smoking cessation.
3.0	Gillian Pearce	• Added smoking status and quit date by 15 weeks gestation
3.1	Gillian Pearce Samantha Rowe TDA	• Tobacco Dependency Advisor provides smoking cessation advice. Systems and processes updated p 6 to 7 and appendix 3 added

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Aim/Purpose

This guideline is for all clinical staff in Maternity.

Introduction

Carbon monoxide (CO) is a colourless, odourless, tasteless and poisonous gas that can kill people. CO is produced when tobacco products are burnt. It is found in inhaled, exhaled and passive smoke. CO can also be produced by exhaust fumes or emitted from malfunctioning or poorly ventilated fossil or wood fuelled heating and cooking appliances.

Exposure to CO is especially dangerous during pregnancy because it deprives the baby of oxygen, slows its growth and development, and increases the risk of miscarriage, stillbirth and sudden infant death. It is important that all pregnant women/birthing people are tested for carbon monoxide, and it is a part of routine antenatal care. A CO level of 4 parts per million or higher is indicative that further investigation or support is required.

This guidance has been written to:

- Identify the procedure for the assessment and safe delivery of CO monitoring at face-to-face antenatal appointments, admissions and in the postnatal period.
- Outline the process for referral to CNWL for smoking cessation.
- Outline the process for follow-up if DNA for CNWL smoking cessation appointment(s).
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The Guidance

Smoking status

All pregnant women /birthing people are assessed at booking for their current and recent smoking status. The woman/ birthing person is given very brief advice about smoking and its impact on the health and wellbeing for her and the baby. Women / birthing people who have a CO reading of 4ppm or more are referred to the smoking cessation service within 2 working days. A quit date is discussed and agreed before 15 weeks gestation. This is documented in the electronic patient record.

Carbon Monoxide Monitoring

This is offered to all women/birthing people at booking and the result documented in the paper and electronic health records. It is repeated at 36 weeks gestation. The pregnant women / birthing people who smoke or have recently quit will have their CO readings done at each antenatal appointment. The women/birthing people are informed about the reason for CO monitoring. Information for women/birthing people can be found at https://smokefreeaction.org.uk/test-your-breath-info and the information is available in other languages.

CO monitoring is not an aerosol generating procedure.
If CO monitoring is declined this is documented on Electronic Patient Record (EPR).

Undertaking CO Monitoring (Appendix 1)

- Hands are cleaned before and after the procedure.
- Attach the disposable mouthpiece to the monitor breath sampling D piece (Pico Smokylzer).
- Turn the monitor on by pressing the power button once.
- Hand the monitor to the woman/birthing person. Ask them to inhale and hold the breath for 15 seconds, the machine will beep when 15 seconds are up.

- The clinician stands 2 metres away from the woman/birthing person during the procedure.
- The woman/birthing person then blows slowly into the device, emptying their lungs completely facing away from the health professional and then hands the device back to the clinician.
- The clinician reads the screen result.
- The woman/birthing person is asked to dispose of the mouthpiece into the hazardous waste bin.
- Remove the D piece to purge the machine with fresh air.
- Clean the machine with a Clinell® wipe.

Results

Changing breath testing thresholds

Traffic light colour	Description	Reading (ppm)	
		piCO™ & Micro+™	piCO baby™
Green	Non-smoker	0-6	0-3
Amber	Borderline	7-9	4-6
1 Red	Smoker – low addicted	10-15	7-10
2 Red	Smoker – moderately addicted	16-25	11-15
3 Red	Smoker – heavily addicted	26-35	16-25
3 Red Flashing	Smoker – very heavy addicted	36+	26+

This table shows the pre-set breath testing thresholds.

See Appendix 2

Women/birthing people with CO readings of 4ppm or above

If the woman/birthing person has a result of 4ppm or above then the clinician must discuss potential causes.

If the pregnant woman/birthing person smokes the clinician will need to explain that this level of carbon monoxide is potentially harmful to their baby's health and wellbeing.

If the pregnant woman/birthing person says that they **have stopped smoking (not recently) or does not smoke** but the CO reading is above 4ppm, then there are other possible reasons for this high reading, and you should explain to them that:

- they may have been exposed to carbon monoxide fumes from a faulty gas boiler, cooker, car exhaust or from paint stripper; the **Gas Safety Advice Line number 0800 300 363** should be given to them at this point.
- they may be lactose intolerant and the high reading is a consequence of consuming dairy products, which can produce gases in their breath.
- exposure to second-hand smoke will not significantly raise a CO result.
- smoking Cannabis or Shisha (Hookah) can elevate CO monitoring.

It is possible that the woman/birthing person is a current smoker but is reluctant to admit this; and so, any further questions should be phrased sensitively.

The impact of smoking and exposure to second-hand smoke in pregnancy to the baby

Health Effect	Maternal Smoking	Second hand smoke
Low birth weight	Double the likelihood	Increased risk
Stillbirth	Double the likelihood	Increased risk
Miscarriage	24-32% more likely	Possible increase
Preterm Birth	27% more likely	Increased risk
Heart defects	50% more likely	Increased risk
Sudden Infant Death	2-3 times the risk	Increased risk
Neonatal death and admissions	Increased risk	Increased risk
Behavioural and Learning Problems	Increased risk	Increased risk
Respiratory Problems	Increased risk	Increased risk

Women/birthing people are referred to smoking cessation service and this includes those who have recently quit smoking in the last 2 weeks. They are to be referred as below:

If you need to refer someone to the Smoking Cessation cervices (ARCH) please email:

cnw-tr.stopsmoking@nhs.net

and please include:

Name of the patient

DOB

NHS NUMBER

Tobacco Dependency Advisor (TDA)

The Tobacco Dependency Advisor is employed by Central and North West London NHS Foundation Trust to provide smoking cessation support at the Trust.

All women/birthing people with a CO reading of 4 or more are referred by the booking midwife to the smoking cessation service. This will allow the TDA to explore if there are any other risk factors. The failsafe is a review of all booking women on a monthly basis and following up women who have a CO reading of 4ppm or more.

The TDA will telephone the women.

Pregnant women / birthing people and Smoking Cessation

The women are invited in for an appointment.

The smoking cessation scheme requires women to stop smoking entirely. Women have to give a quit date. A quit date is discussed and they are followed up weekly for a month and then monthly by the TDA. A CO reading is done at each appointment.

There is an incentive scheme for smoking cessation. The woman / birthing person is given a Love to Shop Voucher for each attendance that she has CO reading below 4ppm. The woman / birthing person can have 1 relapse otherwise she is no longer eligible for the scheme.

Smoking cessation outcomes are monitored by the Trust and the Sector on the National Smoke-free Pregnancy Incentive Scheme digital platform

The TDA documents the contacts and outcomes with the woman / birthing person on the electronic patient record.

Women / birthing people who opt out

If a woman/ birthing person states that she has quit smoking then she needs to have a confirmation CO reading done at the next appointment to verify this. The TDA offers relapse prevention support. The referring midwife / midwifery team is informed of the outcome and next steps.

If the CO reading is below 4 at the next antenatal appointment then continue CO monitoring at each antenatal visit. If the CO reading is above 4 then the midwife needs to discuss the reasons for the reading. Very brief advice is given. The woman / birthing person is given the option for smoking cessation support and referred as required.

Women / birthing people who are smokers and admitted

Women / birthing people are not allowed to smoke on hospital grounds this is Trust Policy. Therefore, to support smokers, they will be offered Nicotine Replacement Therapy. See Appendix 3. And the CNWL Smoking Cessation Pharmacotherapy Guidelines which is located on the intranet. If women decline NRT, they will not be allowed to smoke on Trust Grounds and they will have to smoke outside the perimeter of the hospital at their own risk, no member of staff will support them in this activity. If women / birthing people leave to smoke from the ward area it is to be documented in the electronic patient record.

Those who Did Not Attend (DNA) their Smoking Cessation Appointment.

The TDA will attempt to contact the woman / birthing person 3 times, each attempt is documented on the electronic patient record. If the TDA is unable to contact the woman / birthing person, the referring midwife or midwifery team is notified.

Clinicians should routinely ask at the next contact following referral if the woman/birthing person attended their Smoking Cessation appointment.

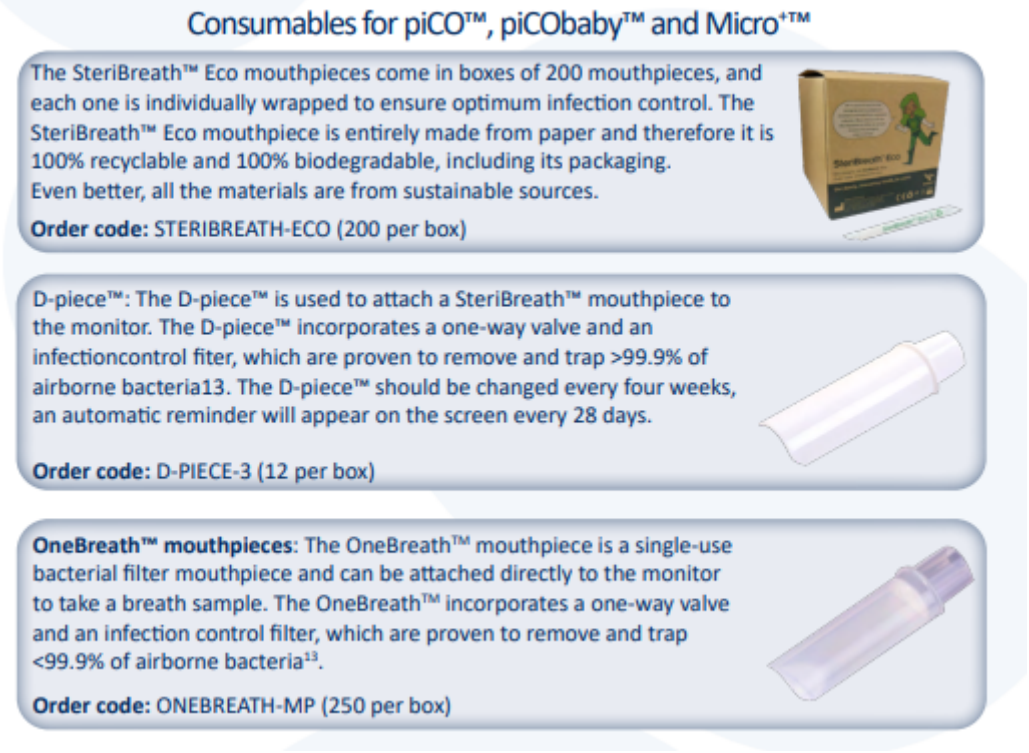
Clinicians need to document on Electronic Patient Record (EPR) if women/birthing people have opted out of smoking cessation services. Those who opt out must be asked at the next appointment if they wish to be referred again. Very brief advice must be given and documented on EPR.

Antenatal CO Monitoring for Smokers/Recent Quitters

All smokers and women/birthing people who have stated that they have quit smoking within the last 2 weeks (of your contact with them) must have their CO levels done at each subsequent antenatal contact; Antenatal Clinic (ANC) appointments, Community Midwife appointments and Day Assessment Unit and Triage encounters.

Care of the Equipment

The D piece is a one-way filter and is replaced every 30 days.



Associated Guidelines, Polices and Documents
Maternal Antenatal Screening Tests including CO monitoring.

Training Need Identified
Annual Mandatory Training for Midwives and MSWs
Very Brief Advice training via the https://portal.e-lfh.org.uk/
CO Monitoring Training
CNWL Ad hoc Training as requested
Saving Babies' Lives Care Bundle Version 3 E-Learning training via iDevelop.

Implementation and Monitoring

All guidelines, protocols and procedures are available on the intranet. All staff have a responsibility to check the Trust's intranet and ensure that they are referring to the most recent issue of the document. The weekly general information bulletin includes notification of publication of new/updated documents.

This clinical guideline, protocol or procedure will be monitored using local and National guidance and via the Trust Incident and Risk Management Policies. If risks/issues are identified further specific monitoring will be considered, for example, clinical audit or risk assessment.

References

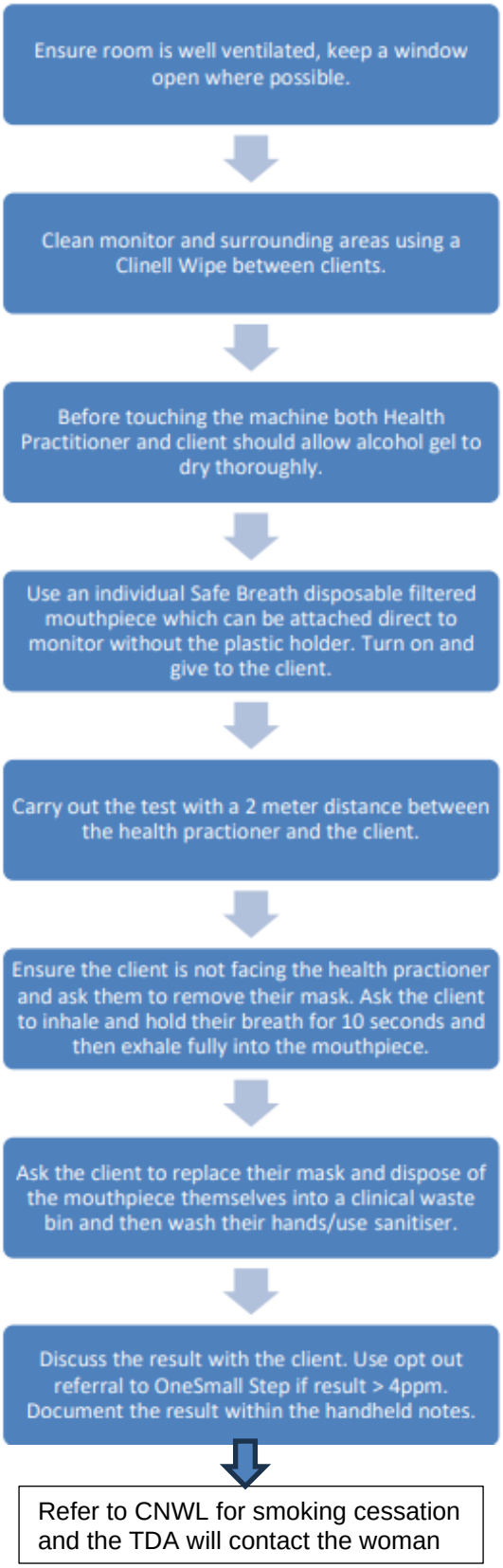
- HSE (2020) Making your workplace Covid- secure during the coronavirus pandemic.
- Public Health England: Guidance on resuming face to face consultations and CO monitoring.
- RCM clinic Briefing Sheet Covid-19
- Saving Babies' Lives Care Bundle Version 3, NHS England (2023)

Appendices

Appendix 1 – Flowchart for performing CO Monitoring
Appendix 2 – What to do with CO Screen Result
Appendix 3 – NRT on Cerner for prescribing

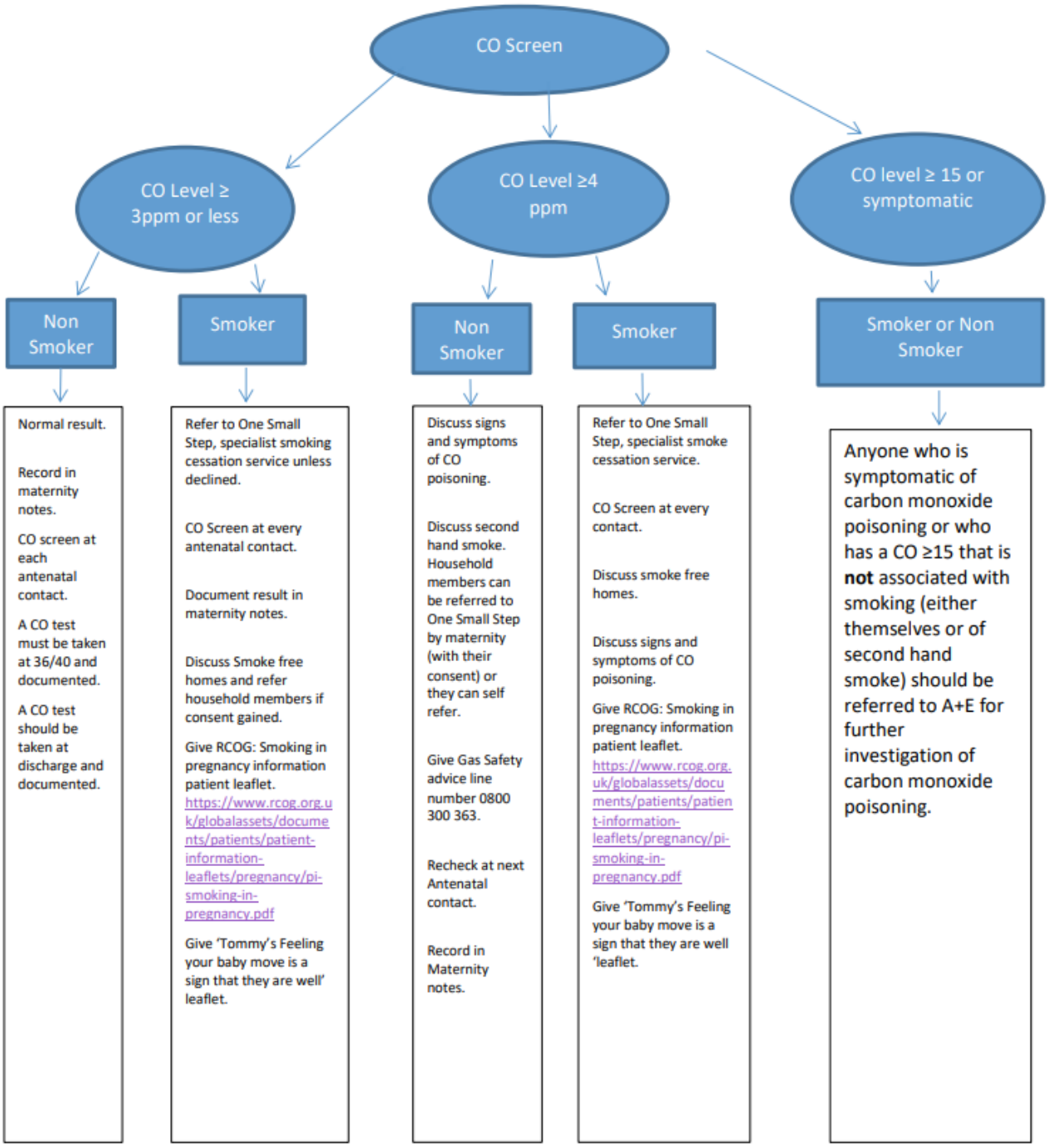
Appendix 1

Flow Chart for performing CO Monitoring



Appendix 2

What to do with a CO Screen result.



Women should be asked about their smoking status at every contact or admission and this should be recorded in their antenatal notes.

N.B. Women/birthing people who have quit within the last 2 weeks should follow the 'Smoker' pathway in the above flowchart.

Appendix 3

NRT prescribing in house

Nicotine Replacement Therapy (NRT), Nicotine Replacement Therapy (NRT) (Initiated Pending)	
FIRST LINE NRT REGIMEN	
On admission, inititate all new patients requiring NRT on the first-line regimen. NRT can be adjusted according to patient response. **Select the 25 mg/16 hour NRT patch AND one short-acting therapy option (according to patient preference), from the first line regimen.**	
Patch	
nicotine (nicotine 25 mg/16 hours transdermal modified-release patch)	1 patch(es), transdermal, once a day, Nicorette Invisi, apply patch to dry non-hairy skin on the hip, trunk or upper arm, remove after 16 hours
nicotine (remove nicotine patch)	REMOVE PATCH, topical, every twenty four hours, Start date: T+1;0000, remove nicotine patch 16 hours after application
Select one short acting therapy;	
Inhalator	
nicotine (nicotine 15 mg inhalation cartridges with device)	1 inhalation(s), inhaled, as required, PRN for nicotine withdrawal, Nicorette
Mouth spray	
nicotine (nicotine 1 mg/dose oromucosal spray sugar free)	1-2 spray(s), oromucosal, as required, PRN for nicotine withdrawal, Nicorette QuickMist
Lozenge	
nicotine	2 mg, oral, lozenge, as required, PRN for nicotine withdrawal, Nicotinell
OTHER LONG-ACTING THERAPY (If alternative to first line therapy required)	
Patch	
16-hour Patches	
nicotine (nicotine 10 mg/16 hours transdermal modified-release patch)	1 patch(es), transdermal, once a day, Nicorette Invisi, apply patch to dry non-hairy skin on the hip, trunk or upper arm, remove after 16 hours
nicotine (remove nicotine patch)	REMOVE PATCH, topical, every twenty four hours, Start date: T+1;0000, remove nicotine patch 16 hours after application
nicotine (nicotine 15 mg/16 hours transdermal modified-release patch)	1 patch(es), transdermal, once a day, Nicorette Invisi, apply patch to dry non-hairy skin on the hip, trunk or upper arm, remove after 16 hours
nicotine (remove nicotine patch)	REMOVE PATCH, topical, every twenty four hours, Start date: T+1;0000, remove nicotine patch 16 hours after application
24-hour Patches	